

Selank Available for Research Use Only

(300 – 900 mcg daily, nasal spray, 2 – 4 weeks on / 2 – 8 weeks off)

Selank is used clinically in Russia via inter-nasal application, as a short-term anxiolytic and cognitive-support peptide, and is reported to improve anxiety, stress resilience, attention, and aspects of working memory without the stimulant-type anxiety typical of classic psychostimulants. It has a good short term safety record, but its lack of long term safety data, combined with its mechanistic biology suggests aggressive cycling is required.

💉💉 Probably safe (some human evidence)

★ ★ Probably effective (strong animal signals and/or some positive human signals)

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.



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Selank is a synthetic heptapeptide (Thr-Lys-Pro-Arg-Pro-Gly-Pro) developed in Russia as an intranasal anxiolytic/nootropic with **immunomodulatory and neuroprotective** actions; clinical and preclinical literature supports **anxiolysis, cognitive stabilization, and stress-buffering**, while community (FRU) reports describe variable nootropic and sleep effects.

The evidence for Selank's long term human safety is moderate because **small controlled human trials and supportive mechanistic and animal data indicate good short-term tolerability**, but single-center concentration and lack of long-term, multi-center safety surveillance reduce confidence.

The evidence for Selank's efficacy for improving cognitive stress response is good because **randomized and controlled human studies plus consistent mechanistic and animal evidence produce a robust convergent signal**, though larger independent trials would be required to move beyond this conservative top cap.

The evidence for Selank's efficacy for improving calm focus is good because **human trials and mechanistic data indicate improved attention and antiasthenic effects without sedation, but heterogeneous endpoints and limited replication weaken the claim.**

Dose and Patterns

Intranasal Selank provides rapid CNS access via olfactory/trigeminal pathways with **subjective effects often reported within 15–60 minutes**. Systemic/plasma peptide levels are

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short-lived due to peptidase degradation, but **central functional effects can persist for several hours**; formulation and amidation materially affect stability and dosing frequency.

Dosing is goal-dependent. Typical practice ranges and cycles are **mechanism-informed and practice-derived**, not universally validated in large randomized trials. Intranasal delivery gives rapid CNS access, so dosing frequency and cycle length balance **acute neurotransmitter effects** with **slower trophic/immune changes** and the risk of adaptive tolerance.

Commonly practice-reported regimens, not validated in large randomized trials are

Goal, Per-administration dose, Daily total (practice-reported), Frequency

- **Situational stress:** 150–300 mcg per dose, 1–3×/day, not to exceed 150–900 mcg daily
- **Daily anxiolysis:** 150–300 mcg per dose, 2×/day, not to exceed 300–900 mcg daily
- **Cognitive enhancement:** 150–450 mcg per dose, 1–2×/day, not to exceed 300–900 mcg daily

Taken together, **never exceed 450 mcg in a single dose**, and **never exceed 900 mcg per day**.

In a clinical settings under supervision, larger doses (up to 600 mcg) and larger daily limits (up to 2,700 mcg) are reported.

Selank (and similar neuroactive peptides) produce benefits by **acutely activating receptors and by engaging slower neurotrophic programs**. **Cycling is necessary** to preserve those benefits and **reduce adaptive downregulation** of the very pathways the peptide targets. **Continuous receptor stimulation** can trigger rapid cellular adaptations: receptors become less responsive, are internalized, or are downregulated at the transcriptional level. This reduces efficacy over days to weeks for peptides that act via membrane receptors.

Benefits

- **Improved anxiety control and reduced stress reactivity — rapid anxiolytic effects without sedation.** (*human clinical data / animal testing*).
- **Enhanced cognitive stability and attention — improvements in short-term memory and learning under stress.** (*animal testing / limited human data*).
- **Upregulation of neurotrophic signaling — increased BDNF expression in hippocampus supporting synaptic plasticity.** (*animal testing / mechanistic*).
- **Immunomodulation and antiviral/anti-inflammatory effects — modulates cytokines (eg, IL-6) and innate antiviral signaling** in preclinical and clinical series. (*human clinical data / animal testing*).
- **Prolongation of endogenous enkephalin activity — inhibition of enkephalin-degrading enzymes** may underlie mood/stress benefits. (*animal testing / mechanistic*).

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- **Consistent individual reports (FRU): improved calm, focus, and sleep regulation** in many users; **some report transient activation or sleep disruption** when dosed late. (*consistent individual reports*).

Indications

- **Generalized anxiety disorder or situational anxiety**
- **Adjunctive therapy for stress-related cognitive impairment or post-stroke rehabilitation**
- **Short-term nootropic use for attention/focus under stress**

Contraindications

- **Pregnancy and breastfeeding**
- **Unstable psychiatric conditions (psychosis, bipolar mania) — risk of destabilizing mood** via monoaminergic modulation
- **Concurrent use of potent CNS depressants or complex polypharmacy without supervision — unknown interactions**

Side effects

- **Transient nasal irritation or burning** after intranasal dosing.
- **Sleep disturbance or vivid dreams when dosed late — dose timing matters.**
- **Occasional transient anxiety or activation in low-stress individuals — baseline stress state influences outcome.**
- **Rare cardiovascular changes reported in preclinical screens — monitor if hypertensive.**

Drivers of Diminished Response and Mitigation Strategies

For **Receptor desensitization** the risk is low to moderate because Selank modulates neuropeptide and GABAergic systems rather than chronically activating a single receptor, as described in available mechanistic studies. For **Neutralizing antibodies** the risk is low because small endogenous-like peptides show low immunogenicity in human studies. For **downstream pathway adaptation** repeated modulation of neurotransmitter systems can alter intracellular signaling and reduce effect. For **System Level Physiological Adaptation** neural homeostatic mechanisms can attenuate benefit. **Overall likelihood of waning is low to moderate.**

Mitigation is **cycling On-Off dosing**.

Monitoring

Monitor mood, sleep, and blood pressure weekly. **Stop and seek review** for new mania, psychosis, sustained hypertension, severe insomnia, or paradoxical anxiety.

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Biological mechanism

Selank is a tuftsin-derived heptapeptide that, when administered **intranasally**, reaches the CNS via **olfactory/trigeminal pathways** and exerts multimodal actions. **Selank modulates GABAergic neurotransmission** (acting as a **positive allosteric modulator** in some models), which contributes to its **anxiolytic profile without sedation**. It **inhibits enkephalin-degrading enzymes**, thereby **prolonging endogenous opioid peptide signaling** and supporting emotional regulation. Selank also **upregulates neurotrophic factors such as BDNF** in the hippocampus, promoting **synaptic plasticity and memory consolidation**. Additionally, Selank **modulates monoamine metabolism (serotonin, dopamine, norepinephrine)** and **affects cytokine balance (eg, IL-6)**, producing combined **neuroprotective, immunomodulatory, and stress-buffering effects** that explain both rapid behavioral changes and longer-term genomic responses observed in animal models.

Stacking Selank and Semax

Combining **Selank** and **Semax** in the same cycle is a common practice in clinics and community protocols because the two peptides have **complementary mechanisms** (Selank: anxiolytic/immune/GABA-monoamine modulation; Semax: neurotrophic/monoamine/melanocortin effects). That complementarity can be useful, but it also raises **additive pharmacodynamic effects** on monoamine systems, sleep, and mood that require careful planning, conservative dosing, and active monitoring. Below I summarize the mechanistic rationale, likely benefits, risks, practical sequencing strategies, monitoring, and example pattern templates — all framed as **practice-derived** options rather than evidence-proven prescriptions.

People combine them because Selank tends to produce rapid anxiolytic and stress-buffering effects via tuftsin-like immune modulation, GABAergic influence, and monoamine balance. While **Semax** emphasizes **BDNF/NGF upregulation** and neuroprotective/memory-enhancing effects while also modulating monoamines. Selank can reduce anxiety and improve stress resilience, which may **unmask or enhance cognitive gains** from Semax (better attention, learning). Semax's trophic effects can consolidate gains made during Selank-mediated windows of lower anxiety and improved focus.

Potential benefits of combining

- **Faster functional gains with better tolerability.** Selank's anxiolysis can make Semax's cognitive effects more usable in real-world tasks (study, rehab).
- **Broader therapeutic coverage.** You address both **acute neurotransmitter balance** (Selank) and **longer-term neuroplasticity** (Semax).

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- **Lower single-agent doses possible.** In some practice patterns, modest doses of each are used together rather than pushing one peptide to a high dose, potentially reducing side-effect risk.

Key risks and interactions to watch when combining

- **Additive monoaminergic effects.** Both influence serotonin/dopamine systems; combined use increases the theoretical risk of **mood destabilization, agitation, or sleep disruption**, especially in people with bipolar disorder or on serotonergic drugs.
- **Sleep disturbance and vivid dreams.** Both can affect sleep architecture; combined late-day dosing increases insomnia risk.
- **Blood pressure effects.** Semax (and some analogs) can transiently affect BP; monitor if cardiovascular risk exists.
- **Unknown long-term safety.** Long-term combined use lacks robust trial data; cumulative exposure should be minimized and periodically reassessed.
- **Formulation and purity variability.** Different suppliers and amidated vs non-amidated variants change potency and stability; mixing formulations increases uncertainty.

Practical combination strategies (mechanism-informed, practice-derived)

1. Sequential blocks (lowest interaction risk)

- **How:** Run a Semax block (focus/neuroplasticity) and follow with a Selank block (mood/stress consolidation), or vice versa.
- **When to use:** If you want to minimize simultaneous monoaminergic load or if psychiatric history is uncertain.
- **Example:** 4–6 weeks Semax on → 2–4 weeks off → 2–4 weeks Selank on → 4–8 weeks off.

2. Concurrent low-dose approach (balanced synergy)

- **How:** Use **lower practice doses** of each peptide simultaneously to gain complementary effects while limiting peak exposure.
- **When to use:** For people with prior tolerance to one peptide, stable mood history, and close monitoring.
- **Practical notes:** Divide daytime doses (morning and early afternoon), avoid evening dosing, and keep total daily exposure conservative.

3. Alternating-day schedule (reduced overlap)

- **How:** Administer Semax on one day and Selank the next (or Semax mornings / Selank afternoons on alternating days).
- **When to use:** If you want some concurrent benefit but reduce same-day additive effects.

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- **Practical notes:** This preserves some continuity while lowering peak combined pharmacodynamic load.

4. Stacked for specific sessions

- **How:** Use Selank on days with high stress or social exposure and Semax on intensive learning/rehab days; avoid using both on the same late-day session.
- **When to use:** For situational optimization (exams, presentations, therapy sessions).

Practical guidance

- **Start conservative.** If combining, begin with **lower doses** and short on-blocks to assess tolerance.
- **Avoid late-day dosing.** Both peptides can affect sleep; schedule doses earlier in the day.
- **Prefer sequencing if psychiatric risk exists.** Sequential blocks reduce simultaneous monoaminergic load.
- **Document effects.** Keep a simple daily log of mood, sleep, BP (if relevant), and cognitive performance to guide adjustments.
- **Always involve a clinician** for anyone with psychiatric history, cardiovascular risk, or concurrent psychoactive medications.

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